

Reza Rahpeima

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Education

2022-present **PhD in Biomedical Electronics and Bioinformatics**

Graduate Institute of Biomedical Electronics and Bioinformatics, College of Electrical Engineering & Computer Science, National Taiwan University, Taipei, Taiwan. GPA: 4.08/4.3.

Thesis: High-fidelity multiphysics simulation engines for physics-informed machine learning in quantitative biomedical imaging and elastography.

Supervisor: Prof. Pai-Chi Li.

2015-2018 **Master of Science in Aerospace Engineering**

Propulsion Department, Faculty of Aerospace Engineering, K. N. Toosi University of Technology, Tehran, Iran. GPA: 17.6/20.

- Ranked 1st in the Department.

Thesis: Numerical analysis of the effect of the stack and its characteristics on the performance of a standing wave thermoacoustic refrigerator.

Supervisor: Prof. Reza Ebrahimi.

2010-2015 **Bachelor of Science in Mechanical Engineering**

Department of Mechanical Engineering, Faculty of Engineering, Bu-Ali Sina University, Hamedan, Iran.

Thesis: Evaluation of the effect of inner and outer transverse and longitudinal grooves on energy absorption characteristics of cylindrical thin-walled tubes under quasi-static axial load.

Supervisor: Prof. Ali Alavi Nia.

2006-2010 **Diploma in Physics and Mathematics Discipline**

Allameh Helli High School, Hamedan, Iran, GPA: 18.89/20.

Affiliated with the National Organization for the Development of Exceptional Talents (NODET).

Research Interests

- Acoustics & Ultrasounds
- Thermoacoustic/Photoacoustic Imaging
- Biomedical engineering
- Heat Transfer and Fluid Dynamics
- Numerical Simulation (CFD, FEM, FVM, FDTD)
- Physics-Informed Machine Learning (PINN)

Honors and Awards

- Awarded Taiwan's National Science and Technology Council (NSTC) Graduate Research Fellowship for outstanding research conducted during my PhD program.
- Awarded a Taiwan Employment Gold Card in the field of Science and Technology.
- Accepted into the exchange program at the "National University of Singapore" and granted the "NTU Voyage of Aspirations" scholarship to cover my expenses throughout my time there."

- Awarded the National Taiwan University Outstanding International Graduate Student Scholarship for pursuing my PhD.
- Recognized as an elite student and permitted to use exceptional talents quota to continue education to Ph.D. at KNTU without participating in the nationwide university entrance exam.
- Ranked 1st in cumulative GPA among students of the propulsion department, class of 2015, K. N. Toosi University of Technology.
- Received the title of the best national thesis in aerospace engineering from the Iranian Society of Aerospace in 2019.
- Received the highest "Scientific Contribution Incentive Award" from K. N. Toosi University of Technology, 2018.
- Iran government higher education bursary during BSc and MSc degrees, 2010 to 2018.
- Recognized as a talented student and awarded membership to the NODET (National Organization for Development of Exceptional Talents) for middle and high school, 2003 to 2010.
- Receiving the certificate of appreciation for attending as executive team at the 16th International Conference of the Iranian Aerospace Society, Tehran, Iran, 2017.

Exams

- **TOEFL: 110**
Reading: 29, Listening: 30, Speaking: 25, Writing: 26

Publications

Publications in refereed Journals:

- **R. Rahpeima**, Chieh-Hsun Wen, Pai-Chi Li, "[A computationally efficient FEM platform for comprehensive simulations of photoacoustic imaging](#)", *Computer Methods and Programs in Biomedicine*, 2025.
- **R. Rahpeima**, Chao-An Lin, "[A comprehensive numerical procedure for high-intensity focused ultrasound ablation of breast tumour on an anatomically realistic breast phantom](#)", *Plos One*, 2024.
- **R. Rahpeima**, I. Nateghi, A. Alavi Nia "[Experimental and numerical investigation of the influence of internal and external corrugations with different geometries on the energy absorption characteristics of cylindrical thin-walled concertina tubes under quasi-static axial load](#)", *Mechanics of Advanced Materials and Structures*, pp. 1-15, 2023.
- **R. Rahpeima**, Chao-An Lin, "[Numerical study of magnetic hyperthermia ablation of breast tumor on anatomically realistic breast phantom](#)", *Plos One*, 2022.
- **R. Rahpeima**, R. Ebrahimi "[A numerical approach for optimization of the working fluid of a standing-wave thermo-acoustic refrigerator](#)", *Engineering with computers*, 2022.
- **R. Rahpeima**, M. Soltani, FM Kashkooli, "[Numerical Study of Microwave Induced Thermoacoustic Imaging for Initial Detection of Cancer of Breast on Anatomically Realistic Breast Phantom](#)", *Computer Methods and Programs in Biomedicine*, 2020.
- M. Soltani, **R. Rahpeima**, FM. Kashkooli, "[Breast cancer diagnosis with a microwave thermoacoustic imaging technique_a numerical approach](#)", *Medical & Biological Engineering & Computing*, pp 1-17, 2019.
- **R. Rahpeima**, R. Ebrahimi, "[Numerical investigation of the effect of stack geometrical parameters and thermo-physical properties on performance of a standing wave thermoacoustic refrigerator](#)", *Applied Thermal Engineering*, pp 1203-1214, 2019.
- A. Alavi Nia, **R. Rahpeima**, S. Chahardoli, and I. Nateghi, "[Evaluation of the effect of inner and outer transverse and longitudinal grooves on energy absorption characteristics of cylindrical thin-walled tubes under quasi-static axial load](#)", *International Journal of Crashworthiness*, pp. 1-12, 2017.

- M. Soltani, **R. Rahpeima**, F. Moradi Kashkooli, A. Alipoor, and P. Torkaman, "Numerical modeling of breast cancer diagnosis with microwave thermo-acoustic imaging", *Journal of Modares Mechanical Engineering (in Farsi)*, pp. 425-433, 2018.

Conference Presentations:

- **R. Rahpeima**, Pai-Chi Li, "A comprehensive FEA platform for realistic simulations of photoacoustic breast imaging", *IEEE International Ultrasonic Symposium 2023*, Montreal, Canada (Poster Presentation).
- **R. Rahpeima**, Chieh-Hsun Wen, Jian-yu Lu, Pai-Chi Li, "Super-Resolution Photoacoustic Imaging by Modulation of the Point Spread Function", *IEEE International Ultrasonic Symposium 2025*, Utrecht, Netherlands (Oral Presentation).
- **R. Rahpeima**, Ting-Wei Chen, Pai-Chi Li, "A Comprehensive Physics-Based Finite-Element Framework for End-to-End Shear Wave Elastography Simulation: the Cornea As a Test Case", *IEEE International Ultrasonic Symposium 2026*, Raleigh, North Carolina, USA (Oral Presentation).

Publications Under Review:

- **R. Rahpeima**, Chieh-Hsun Wen, Jian-yu Lu, Pai-Chi Li, "Super-Resolution Photoacoustic Imaging by Modulation of the Point Spread Function", *Photoacoustics*, 2026.
- **R. Rahpeima**, Ting-Wei Chen, Pai-Chi Li, "An end-to-end physics-based finite-element framework for simulation of acoustic radiation force-induced shear wave elastography: human cornea as a test case", *Computer Methods in Applied Mechanics and Engineering*, 2026.

[Google Scholar](#)
[Website](#)

Research Projects

- Development of inverse physics-informed neural networks (PINNs) for super-resolution **shear wave elastography** and quantitative material-property estimation, 2026-present.
National Taiwan University, Under the supervision of Prof. Pai-Chi Li.
- Computational modeling of acoustic radiation force-induced shear waves in the human cornea and microchannels for advanced **shear wave elastography imaging** applications, 2025-present.
National Taiwan University, Under the supervision of Prof. Pai-Chi Li.
- Investigation of a super-resolution technique through PSF modulation, with applications in **photoacoustic microscopy imaging**, 2023-2025.
National Taiwan University, Under the supervision of Prof. Pai-Chi Li.
- Simulation and experimental study of **photoacoustic imaging** using row-column array beamforming, specifically tailored for this imaging technique, 2022-2023.
National Taiwan University, Under the supervision of Prof. Pai-Chi Li.
- Simulation of **high-intensity focused ultrasound** ablation of breast tumor, 2022-present.
National Tsinghua University, Under the supervision of Prof. Chao-An Lin.
- Simulation of **magnetic fluid hyperthermia** ablation of breast tumor, 2021-2022.
National Tsinghua University, Under the supervision of Prof. Chao-An Lin.
- Simulation of microwave **thermoacoustic tomography** platform and analysis of the results for breast cancer detection in early stages, 2016-2021.
K. N. Toosi University of Technology. Under the supervision of Dr. M. Soltani.
- Simulation and manufacturing of a standing wave **thermoacoustic refrigerator** and evaluating the effect of the stack geometry and the working fluid on the performance of the models, 2015-2020.

K. N. Toosi University of Technology. Under the supervision of Prof. R. Ebrahimi.

- Modeling numerical simulations and developing experimental tests on cylindrical **thin-walled tubes** with inner and outer transverse and longitudinal grooves and also with internal and external waves, 2014-2015.
Bu-Ali Sina University. Under the supervision of Prof. A. Alavi-Nia.

Academic/Work Experience

- **Simulation Engineer (SHL Medical):** Performing plant simulation with "Plant simulation & throughput optimization" Siemens Software and developing Python codes for different data analysis purposes, 2022-2023.
- **Research Assistant of Ultrasonic Imaging laboratory (National Taiwan University):** Conducting research and carrying out research papers in the field of photoacoustic imaging and ultrasound elastography, 2022.
- **Research Assistant of CFD laboratory (National Tsinghua University):** Conducting research and Carrying out research papers in the field of HIFU and MFH ablation, 2021-2022.
- **Research Assistant of Combustion and Propulsion Laboratory (K. N. Toosi University of Technology):** Conducting research and carrying out several research papers in the field of thermoacoustics, 2018-2021.
- **Teaching Assistant of undergraduate courses "Thermodynamics 2" & "Heat Transfer 1" (K. N. Toosi University of Technology):** Preparing and conducting weekly tutorials for a class of 40 students as well as marking duties, 2016-2017.
- **High school teacher (Several high schools in Taipei, New Taipei, and Hsinchu, Taiwan):** Preparing teaching materials and teaching Physics, Math, and English to a class of several high school students, 2022-present.

Community Involvement

- **Scientific peer-reviewer:** reviewer for several journals such as "*IEEE Transactions on Medical Imaging*", "*Medical Physics*" and "*Applied Clinical Medical Physics*", and "*Frontiers in Thermal Engineering*", 2023-present.
- **Conference Assistant:** 16th International Conference of Iranian Aerospace Society, Tehran, Iran, 2017. Assisted as a member of the executive team.
- **Robotic Team Membership:** Participating as a member of the technical group of the Bu-Ali Sina University robotic team, 2013-2015.

Skills

- **Artificial Intelligence and Machine Learning:** +2 years of experience in development of physics-informed neural networks (PINNs) for forward and inverse modeling and image reconstruction executed in the context of **ultrasound elastography** and **biomedical imaging**. Using Python-based frameworks such as **PyTorch** and **TensorFlow**.
- **Physics-Based Computational Modeling (Simulation):** +9 years of experience in numerical methods and simulation in different areas of acoustics, fluid mechanics, heat and mass transfer, electromagnetics, and solid mechanics executed in the context of **Ultrasonounds**, **photo/thermo-acoustics**, **biomedical imaging**, **MEMS**, **medical ablation**, and **thermo-fluids**. Using commercial software such as **COMSOL Multiphysics**, **Ls-Dyna**, **Ansys Fluent**, and **OpenFOAM**.
- **Coding and Programming:** +9 years of experience in writing codes and developing algorithms (Mostly **C++** and **MATLAB** and some **Python** and **Julia**) for various purposes such as **beamforming algorithms**, **signal processing**, **numerical methods** such as FDTD, FEM, FVM, and Lattice-Boltzmann, and **machine learning**.
- **Modeling:** +5 years of experience in modeling complex 3D models (Mostly **Catia** and **SolidWorks** and some **AutoCAD**) such as modeling a realistic **human breast phantom** derived from MRI images of patients.

- **Language:** **Persian** as the native language, fluent in **English**, and basic in **Chinese** and **Arabic**.
- Having the **license of “principles and theory of piping”** from the research training center of Tehran University, 2015.
- Having the **license of “Catia advanced designing”** from the Department of Technical and Vocational Education of Iran, 2012.

References

- *Prof. Pai-Chi Li*, National Taiwan University. paichi@ntu.edu.tw
- *Prof. Chao-An Lin*, National Tsinghua University. calin@pme.nthu.edu.tw
- *Prof. Reza Ebrahimi*, K. N. Toosi University of Technology. rebrahimi@kntu.ac.ir
- *Prof. Madjid Soltani*, K. N. Toosi University of Technology. msoltani@kntu.ac.ir